Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	***AI-Powered Infrastructure for Ultrafast Simulation and Optimization of Complex Fluid Systems***				
Korea's 12 National Strategic Technology Fields	**Semiconductor/Display (), Secondary Battery (), Advanced Mobility ( ), Next-gen Nuclear Power ( ),** **Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI (√),** **Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( )**				
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)			**Technology Field 2** (Large Category)	
	Mechanical・Materials			***Knowledge Services***	
	Medium Category		Small Category	Medium Category	Small Category
	Energy / Environment Mechanical Systems		***Fluid Machinery***	***R&D / Engineering Services***	***Production Process Modeling / Simulation***
Concept of Proposed Technology	***The proposed technology targets the development of an AI-powered simulation infrastructure for ultrafast, high-fidelity computation of complex fluid systems, motivated by the fundamental performance limitations of conventional numerical solvers when applied to multiscale and multi-physics systems****. Many fluid phenomena of scientific and engineering interest exhibit tightly coupled spatial and temporal scales, where resolving fine-scale structures requires both extremely high spatial resolution and prohibitively small time steps, leading to computational costs that scale beyond practical limits for existing CFD pipelines.*				

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	To address these challenges, the proposed infrastructure adopts neural representations and neural PDE solvers as the core computational substrate, enabling a reformulation of fluid simulation that directly targets the dominant sources of numerical cost and error accumulation. Rather than relying on repeated local time stepping and grid-based updates, the proposed approach leverages learned, physics-consistent representations of spatiotemporal dynamics to approximate long-range transport, operator application, and solution evolution in a compact and highly parallelizable form. This design allows the simulation to preserve essential physical structures—such as vortical features, interface geometry, and conserved quantities—while substantially reducing the number of explicit numerical operations required to advance the solution. *The infrastructure integrates three tightly coupled capabilities. (1)* ***Neural Physical Representations****: Continuous neural fields are used to encode velocity, pressure, and auxiliary physical quantities over space and time, providing high-resolution, differentiable representations that avoid excessive discretization while supporting accurate evaluation of differential operators. (2)* ***Neural PDE Acceleration****: Physics-informed and operator-learning-based neural solvers are employed to accelerate key components of the governing equations, including transport, coupling, and constraint enforcement, enabling rapid progression of the simulation state with controlled physical error. (3)* ***Ultrafast GPU-Native Execution:*** *All representations and solvers are designed for GPU-native execution, exploiting massive parallelism to achieve ultrafast simulation performance at resolutions and fidelities that are difficult to attain with traditional methods.*
Rationale for Joint R&D Project	*From the research institute's perspective, the proposed joint R&D is equally well motivated. As a research group in School of Interactive Computing at Georgia Tech that is leading the development of ultrafast and high-fidelity simulation methods for complex fluid systems, we seek to actively engage with industrial partners to translate our advances in neural representations and neural PDE solvers into impactful, real-world applications. Collaboration with industry provides critical opportunities to validate our simulation infrastructure on practical problem settings, identify new performance challenges, and guide future research directions toward high-impact use cases. Through such partnerships, we aim to ensure that our computational innovations are not only scientifically advanced, but also directly relevant to emerging simulation-driven industries.* *Rationale for collaboration:* *(1) The research institute contributes expertise in neural representations, neural PDE solvers, and high-performance fluid simulation, enabling the development of a robust and efficient simulation infrastructure.* *(2) Industrial partners contribute real application scenarios, performance targets, and validation workloads, ensuring that the infrastructure is aligned with practical computational demands.* *(3) Joint co-development is essential to produce a reusable simulation infrastructure that meets industrial requirements for accuracy, speed, and scalability.* *Expected benefits of joint R&D:*

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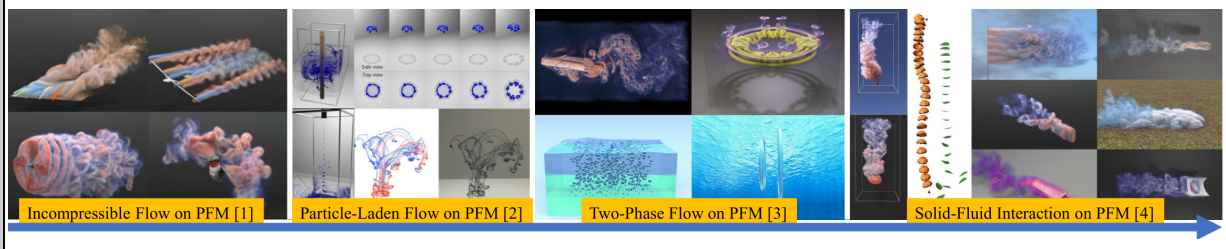
	(1) Reduced computational barriers for startups and SMEs by providing access to a shared, high-performance simulation infrastructure. *(2) Faster translation of advanced simulation methods into practical industrial use through iterative validation.* *(3) Creation of a scalable foundation that strengthens simulation-driven industries and accelerates innovation.*  *Figure. 1 The various high-performance fluid simulation systems PI developed.*
Key Words	*1. Ultrafast Fluid Simulation* *2. High-Fidelity Computational Fluid Dynamics* *3. Neural Representations for Physical Simulation* *4. Neural PDE Solvers* *5. Simulation Infrastructure for Complex Fluid Systems*

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL) [2)]		**Start of the Project**	Level (3)	**End of the Project**	Level (6)
R&D Objectives		*The objective of this R&D project is to develop an AI-powered simulation infrastructure for ultrafast, high-fidelity simulation and optimization of complex fluid systems, enabling substantial performance improvements over conventional CFD methods while preserving physical accuracy and numerical robustness across representative problem classes.* *The specific R&D objectives are:* ● *To develop neural representations for complex fluid states, enabling compact, continuous, and differentiable encoding of velocity, pressure, and auxiliary physical quantities suitable for high-resolution simulation* ● *To develop neural PDE solvers for accelerated fluid dynamics computation, targeting key components such as transport, coupling, and constraint enforcement to significantly reduce computational cost while maintaining controlled physical error* ● *To establish a GPU-native, high-performance simulation infrastructure, capable of delivering ultrafast execution and validated accuracy on representative industrial fluid simulation and optimization tasks*			
R&D Target Specifications		*The target performance specifications of this R&D project are defined in terms of computational speed, simulation fidelity, and scalability, reflecting the core objective of delivering ultrafast and accurate simulation for complex fluid systems.* ***Simulation Performance (two orders of magnitude speedup):*** *Achieve a 10× to 100× speedup over representative conventional CFD solvers for targeted classes of fluid problems, while maintaining stable numerical behavior and consistent physical outputs under comparable resolution and boundary conditions.* ***Simulation Fidelity and Accuracy:*** *Achieve high-fidelity simulation results that preserve key physical structures, including vortical dynamics, interface geometry, and conserved quantities, with quantitative errors maintained within acceptable bounds relative to high-resolution baseline solvers on benchmark problems.* ***Scalability and Resolution Capability****: Demonstrate the ability to simulate high-resolution and multi-scale fluid systems that are computationally impractical for traditional methods, including large spatial domains, long time horizons, or tightly coupled multiphysics scenarios, while sustaining efficient GPU-native execution.*			

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R&D Plan		*This R&D project will be carried out over three years, following a staged development and validation plan that progressively advances the proposed ultrafast simulation infrastructure for complex fluid systems from core algorithmic foundations to validated, industry-relevant computational capabilities.* *Year 1: Core Algorithm Development and Performance-Oriented Prototyping* *The first year will focus on establishing the theoretical and computational foundations of the proposed simulation infrastructure. Key activities include the development of neural representations for fluid states and the design of neural PDE solvers that accelerate critical components of fluid dynamics computation, including transport, coupling, and constraint enforcement. Initial GPU-native implementations will be developed to assess raw performance characteristics, and benchmark studies will be conducted against representative high-fidelity CFD solvers to evaluate speedup, accuracy, and numerical stability on controlled test cases.* *Year 2: Integrated Simulation Infrastructure and Large-Scale Validation* *In the second year, the project will focus on integrating individual algorithmic components into a unified, end-to-end simulation infrastructure. Emphasis will be placed on scalability, robustness, and sustained high-performance execution on modern GPU platforms. The integrated system will be evaluated on larger and more complex fluid simulation problems, including multi-scale and multiphysics scenarios, to validate its ability to deliver high-fidelity results under realistic computational workloads. Comparative studies will be conducted to quantify performance gains and fidelity trade-offs relative to traditional numerical methods.* *Year 3: Application-Driven Evaluation and Technology Transfer Preparation* *The final year will emphasize extensive validation and refinement of the simulation infrastructure through application-driven use cases relevant to industrial and startup environments. The focus will be on demonstrating reliable performance, reproducibility, and scalability for problem settings that are computationally prohibitive for conventional solvers. Based on these results, the infrastructure will be further optimized and documented to support external adoption, and pathways for industrial deployment and technology transfer will be assessed to facilitate the transition of the proposed simulation technology toward broader practical use.*

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	*The research institute will be responsible for the development of core AI algorithms and system-level intelligence for the proposed home mobile manipulation technology. The institute will lead the research on perception, decision-making, and control methodologies required to achieve reliable robot operation in domestic environments.* *The specific roles and responsibilities of the research institute include:* *● Development of AI-based navigation, object search, and mobile manipulation algorithms, including vision-language-based perception and adaptive skill coordination* *● Design of the overall system architecture and integration framework for AI-driven mobile manipulation* *● Simulation-based development, algorithm validation, and performance analysis of the proposed technologies* *Technical guidance and collaboration with the industrial partner to support system integration and experimental evaluation*
Expected R&R of Korean SME Partner	*The Korean SME partner will be responsible for the application-driven integration, validation, and deployment of the proposed ultrafast simulation infrastructure for complex fluid systems. The SME will play a critical role in translating advanced simulation algorithms into practical, reliable, and industry-ready computational capabilities that can be adopted by startups and engineering teams.* *The specific roles and responsibilities of the Korean SME partner include:* *● Provision of industrial simulation use cases and performance requirements, contributing representative problem settings, constraints, and evaluation criteria drawn from real-world engineering and startup-driven applications* *● Integration and validation of the simulation infrastructure, including deployment on industrial computing platforms and evaluation under realistic computational workloads* *● Execution of large-scale benchmarking and performance testing, encompassing speed, scalability, numerical robustness, and accuracy relative to existing industrial CFD tools* *● Contribution to refinement, packaging, and commercialization planning, leveraging industrial expertise to support usability, reliability, and adoption of the simulation infrastructure in commercial and startup environments* *Through these responsibilities, the Korean SME partner will ensure that the proposed simulation infrastructure delivers measurable industrial value and supports practical technology transfer beyond the research setting.*

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	*The proposed joint R&D project is expected to deliver a validated AI-powered simulation infrastructure for ultrafast, high-fidelity modeling and optimization of complex fluid systems. The project will advance the state of the art in neural representations and neural PDE solvers for physical simulation, achieving measurable improvements in computational performance, scalability, and simulation fidelity compared to conventional CFD approaches. The resulting infrastructure will enable fast and accurate simulation of challenging fluid phenomena that are difficult or impractical to address with existing methods.* *From an industrial perspective, the project will provide the Korean SME partner with core computational technologies and performance-validated simulation capabilities that can be directly applied to simulation-driven product development, digital twins, and engineering optimization. The outcomes will lower the computational barrier for startups and SMEs that depend on high-fidelity fluid simulation, supporting faster design cycles, reduced development cost, and more competitive innovation. More broadly, the project is expected to strengthen the simulation and engineering technology ecosystem for Korean SMEs and contribute to the growth of AI-enabled, simulation-driven industries through sustained international collaboration and technology transfer.*

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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